

Economic Analysis of Financial Markets (EAFM)

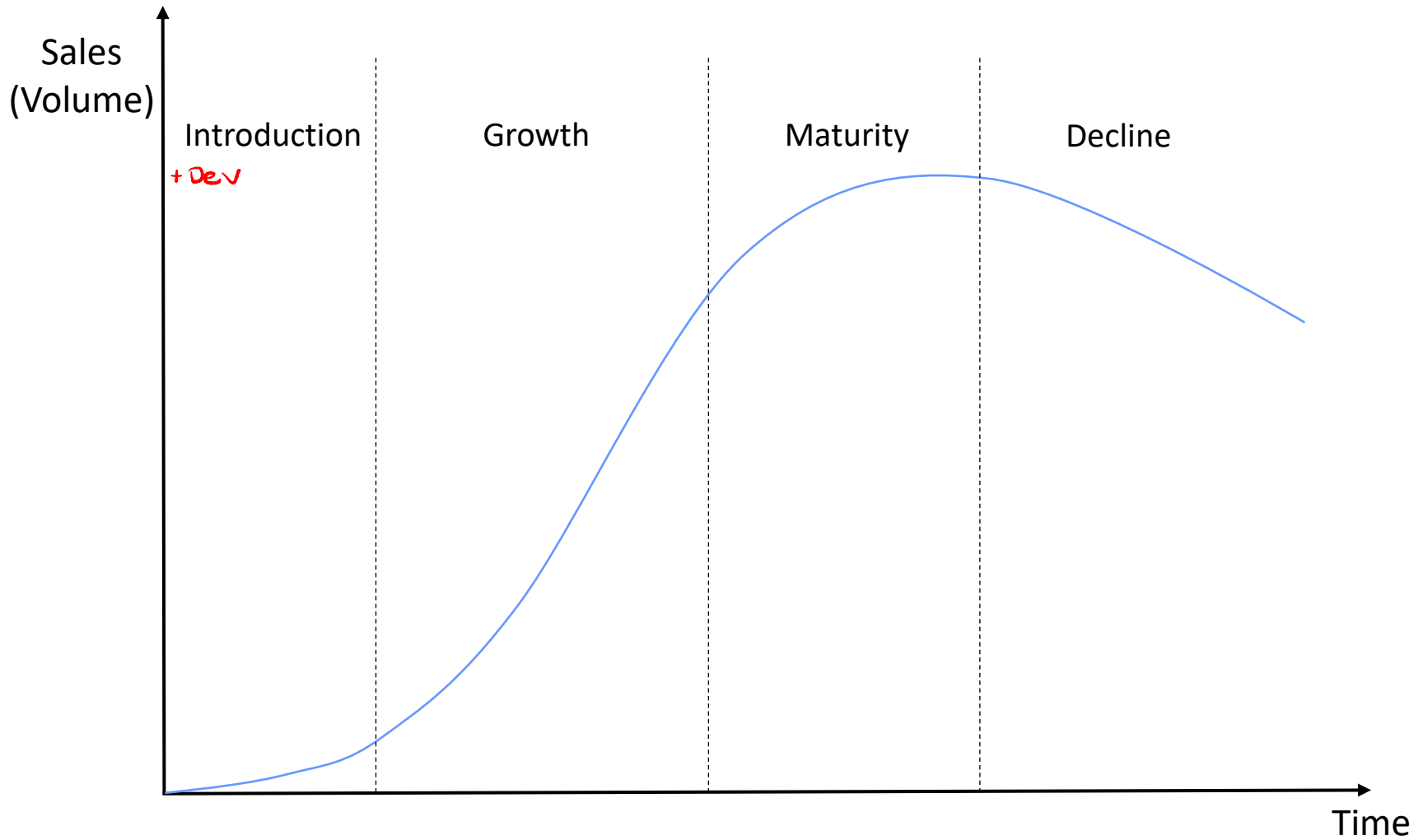
Economics Module

Lecture 2

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Product Life Cycle



Product/Industry Life Cycle

- **Stage 1 - Introduction and Development**

- Product new to the market
- Demand has to be “created”
 - Product complexity
 - Degree of newness
 - Consumer needs
 - Presence of competitive substitutes
- Rapid growth in sales and earnings
- High costs ----> Key driver of almost all failures!
- Second mover advantage?



Product Life Cycle

- **Stage 2 – Growth**

- Gradual rise in sales
- Marked increase in consumer demand
- Performance in industry tracked by performance of surviving firms
- *'Try the product' vs 'prefer the brand'*
- Importance of pricing correctly
- Sold in more locations
- Product improvements
- Note: Factory sales rise more rapidly than store sales (outsiders view as profit)
 - Filling of distribution pipeline
- Growth rate of the industry still faster than the economy



Product Life Cycle

• Stage 3 – Maturity

- Look out for market saturation
- Slow, stable rates of growth
 - Track growth of economy in general
- **Intensity of price competition increases**
 - Increased pressure on profits
- Producer now competes on
 - Price
 - Marginal product differences, or both
 - Packaging and advertising
- Little opportunity for organic growth
- Duration of maturity phase undetermined
 - **Long:** Beer, steel
 - **Short:** Women's fashion fads

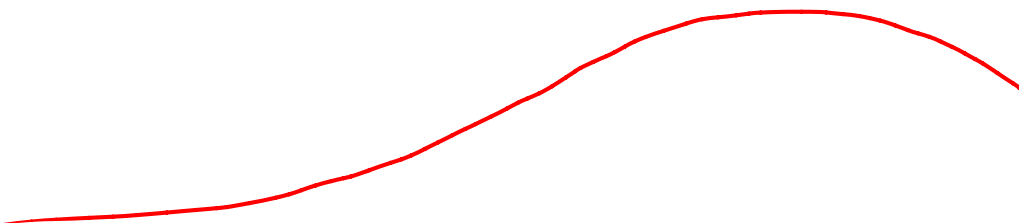


Product Life Cycle

- **Stage 4 – Decline**

- Decline in demand
- Overcapacity a major issue
 - Costs are counter-optimal
- Production concentrated among few producers
- Prices and margins depressed
- Profits become more of a challenge linked to production and distribution efficiency rather than increased sales





Stage Factor	Introduction	Growth	Maturity	Decline
Generic strategies	Differentiation	Differentiation	Differentiation Overall cost leadership	Overall cost leadership Focus
Market growth rate	Low	Very large	Low to moderate	Negative
Number of segments	Very few	Some	Many	Few
Intensity of competition	Low	Increasing	Very intense	Changing
Emphasis on product design	Very high	High	Low to moderate	Low
Emphasis on process design	Low	Low to moderate	High	Low
Major functional area(s) of concern	Research and development	Sales and marketing	Production	General management and finance
Overall objective	Increase market awareness	Create consumer demand	Defend market share and extend product life cycles	Consolidate, maintain, harvest, or exit

Analysing the firm's microeconomic environment

PORTER'S FIVE FORCES

- Tool to analyse the firm's microeconomic environment
- An organisation's operating environment is predominantly conditioned by the intensity of the competition in the industry within which it is competing
 - Influence on competitive rules of the game
 - Strategies available to the firm
- Three horizontal forces
 - Competition from incumbents ^{rivals}
 - Competition from substitute products
 - Competition from new entrants
- Two vertical forces
 - Bargaining power of buyers
 - Bargaining power of suppliers



Analysing the firm's microeconomic environment

- The Firm's Value Chain

- What defines successful firms?

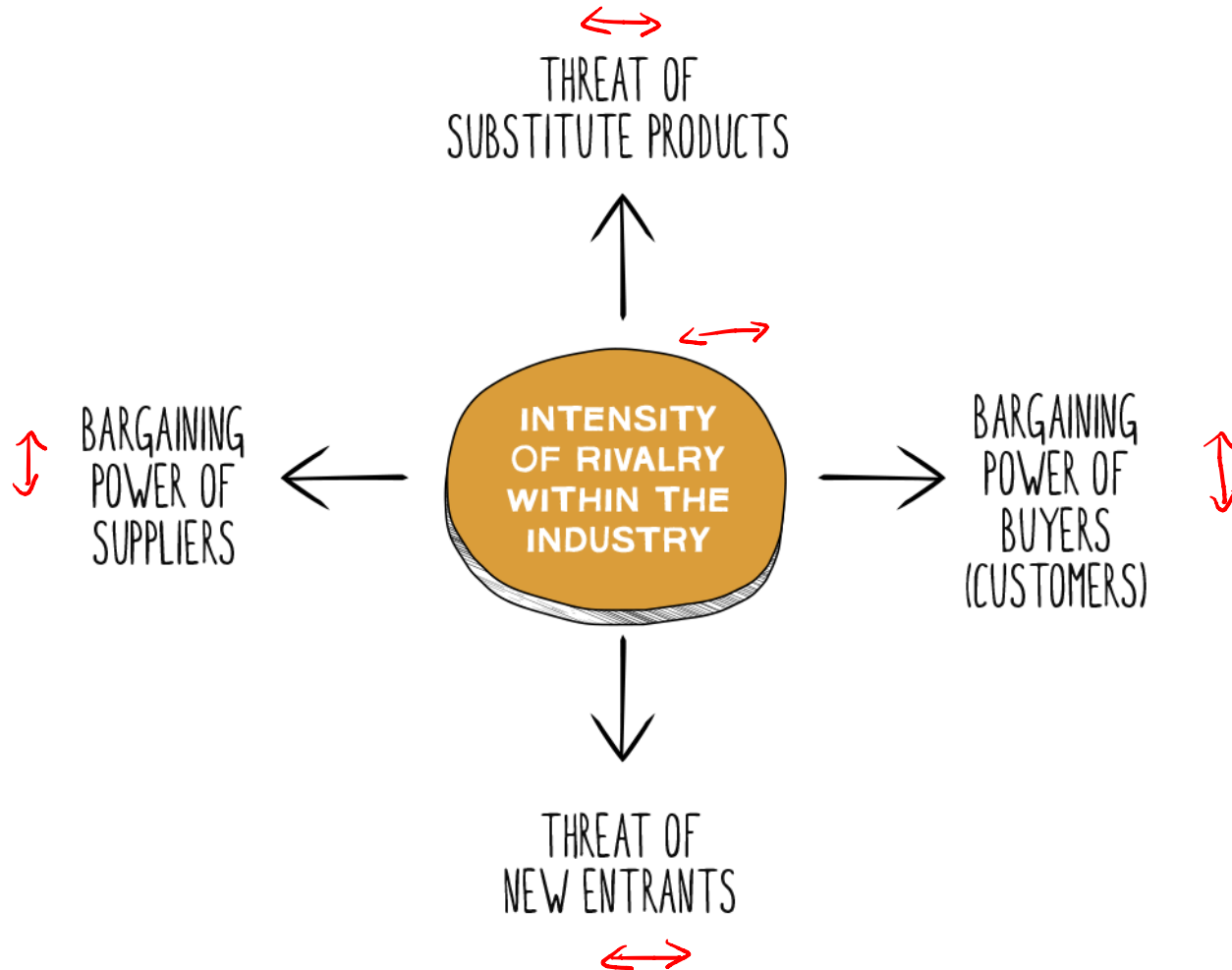
- Sustainable competitive advantage

- Ability to deliver a product at a lower cost than rivals OR

- Create an offering with unique benefits to the buyer that justifies charging a premium price

- Separate firm into the discrete but interrelated activities

PORTER'S FIVE FORCES MODEL



Michael Porter's "Five Forces of Competition" framework describes how the structural features of an industry influence the distribution of value created by firms within that industry.

- Ideally, firms in an industry would like to capture most or all of the economic value that they create.
- However, competitive forces operate to push that value "forward" to customers (in the form of lower prices), or in some cases, "backward" to suppliers.

Competitive Forces Shape Strategy



- The essence of strategy formulation is to better cope with competition.
- Competition in an industry is rooted in its underlying economics, and competitive forces exist that go well beyond the established combatants in a particular industry.
- The corporate strategists' goal is to find a position in the industry where his or her company can best defend itself against these forces or can influence them in its favor.

Industry Analysis & Competitive Analysis



- An **industry** is a collection of firms that offer similar products or services.
- **Structural attributes** are the enduring characteristics that give an industry its distinctive character.
- **Concentration** refers to the extent to which industry sales are dominated by only a few firms.
- **Barriers to entry** are the obstacles that a firm must overcome to enter an industry.

Five Competitive Forces

The Idea in Brief

↔ • Customers

↔ • Suppliers

↔ • Entrants

↑↓ • Substitutes

↑↓ • Competition *incumbents / rivals*

Five Competitive Forces

Threat of new entrants

- If profitability high - attract **new entrants that undercut prices**
research + dev.
- R & D expenditures will increase to keep market share
- High profit margins will not last unless significant barriers to entry
 - Economies of scale (supply side)
 - Economies of scale (demand side)
 - Switching costs
 - Capital requirements
 - Incumbency advantages
 - Distribution channels
 - Government

Threat of entrants

- Entry hurts the incumbents in two different ways
 - Entry cuts into the incumbents' market share
 - Entry intensifies internal rivalry and leads to a decline in price cost margin
- **Implications**
 - Almost like rival producers (when entry is fast)
 - Impeded by “entry barriers” (costs of entry)
 - Incumbents can take actions to deter entry

Five Competitive Forces

The power of suppliers

- Push up prices
- Problem if supplier market more concentrated
- Suppliers not dependent on particular industry
- High switching costs
- Lack of substitutes
- Integration threats

Implications

Supplier Power

- Suppliers can ^{take} siphon value from producers
- Power increases with supplier concentration
- Analysis similar to buyer power

Five Competitive Forces

The power of consumers

BP depends on buyer concentration, info etc

- Force down prices
- Negotiation power
 - Few buyers purchasing large volumes
 - Standardized products
 - Few switching costs
 - Integration threat (backwards)
- Price sensitive
 - Significant cost (relative to budget)
 - Low profit
 - Quality of buyer's product not dependent on inputs

According to Porter (1980), the bargaining power of buyers depends on buyer concentration, information, and other factors.

Implications

Buyer power greater when:

- Buyers are more concentrated
- Buyers are better informed *information asymmetry*

Five Competitive Forces

The threat of substitutes

- Price pressure
- Threat depends on the potential substitute:
 - Price vs Performance (eg. Skype vs Telkom)
 - Switching cost (generic drugs)
 - New technology

Implications

Competition from Substitutes

willingness to pay

- Reduces buyers' WTP for the industry's product.
- Strengthens bargaining position of single buyer.
- Given many buyers with varied WTP, lowers the demand curve for the industry's product.
- If substitute price falls or quality improves, buyer's WTP for the focal industry's product falls.

Five Competitive Forces

Competition

- Intensity of competition depends on:
 - Number and size of firms (HHI, FFCR)
 - Growth rate
 - Exit barriers
 - State owned competitors
- Price competition particularly bad for profits
 - Identical products
 - High fixed costs
 - Perishable products

Implications

Rivalry increases with

- More direct competitors
- Industry excess capacity
- Exit barriers

Extension

Impact of Complements

- Sometimes called the “sixth industry force.”
- Can be viewed as opposite of substitutes.
- *Increases* buyer’s WTP for the industry’s product.
- Raises the demand curve for the industry’s product.
- If complement price falls or quality improves, buyer’s WTP for the industry’s product rises (consider strength of relationship)

Economic Factors



- Concern nature and direction of economy in which a firm operates
- Types of factors
 - General availability of credit
 - Level of disposable income
 - Propensity of people to spend
 - Prime interest rates
 - Inflation rates
 - Trends in growth of gross national product

Social Factors



- Include beliefs, values, opinions, and lifestyles of people
- Recent social trends
 - Entry of large numbers of women into labor market
 - Accelerating interest of consumers and employees in quality-of-life issues
 - Shift in age distribution of population

Political Factors



- Define legal and regulatory parameters within which firms must operate
- Types of factors
 - Fair-trade decisions
 - Competition laws
 - Tax programs
 - Minimum wage legislation
 - Pollution and pricing policies
 - Administrative jawboning

Technological Factors



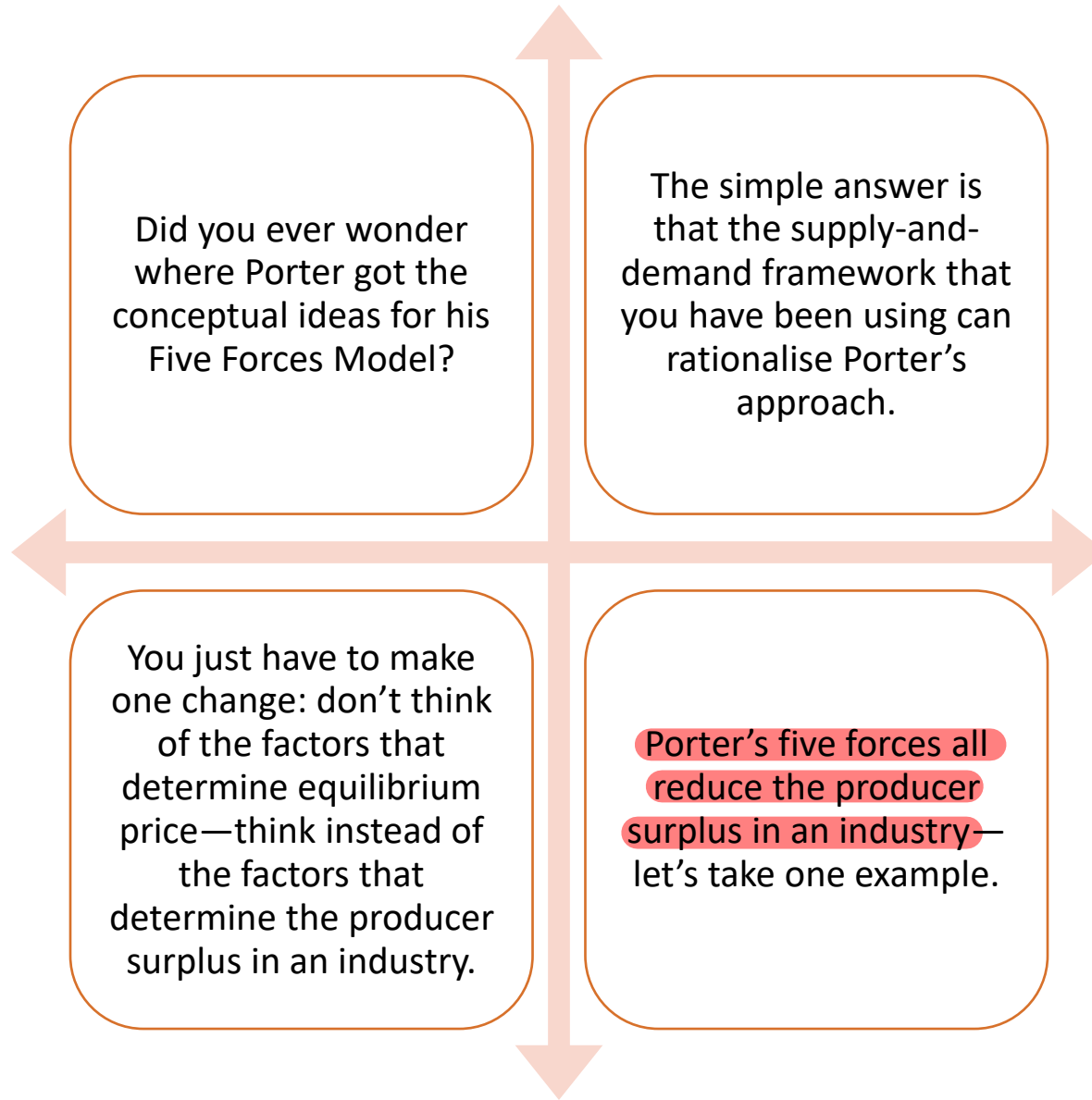
- Focus on technological changes affecting industry
- Types of changes
 - New products
 - Improvements in existing products
 - Manufacturing and marketing techniques
- Role of technological forecasting
 - Foresees advancements and estimating their impact
 - Alerts managers to impending challenges and promising opportunities

Ecological Factors

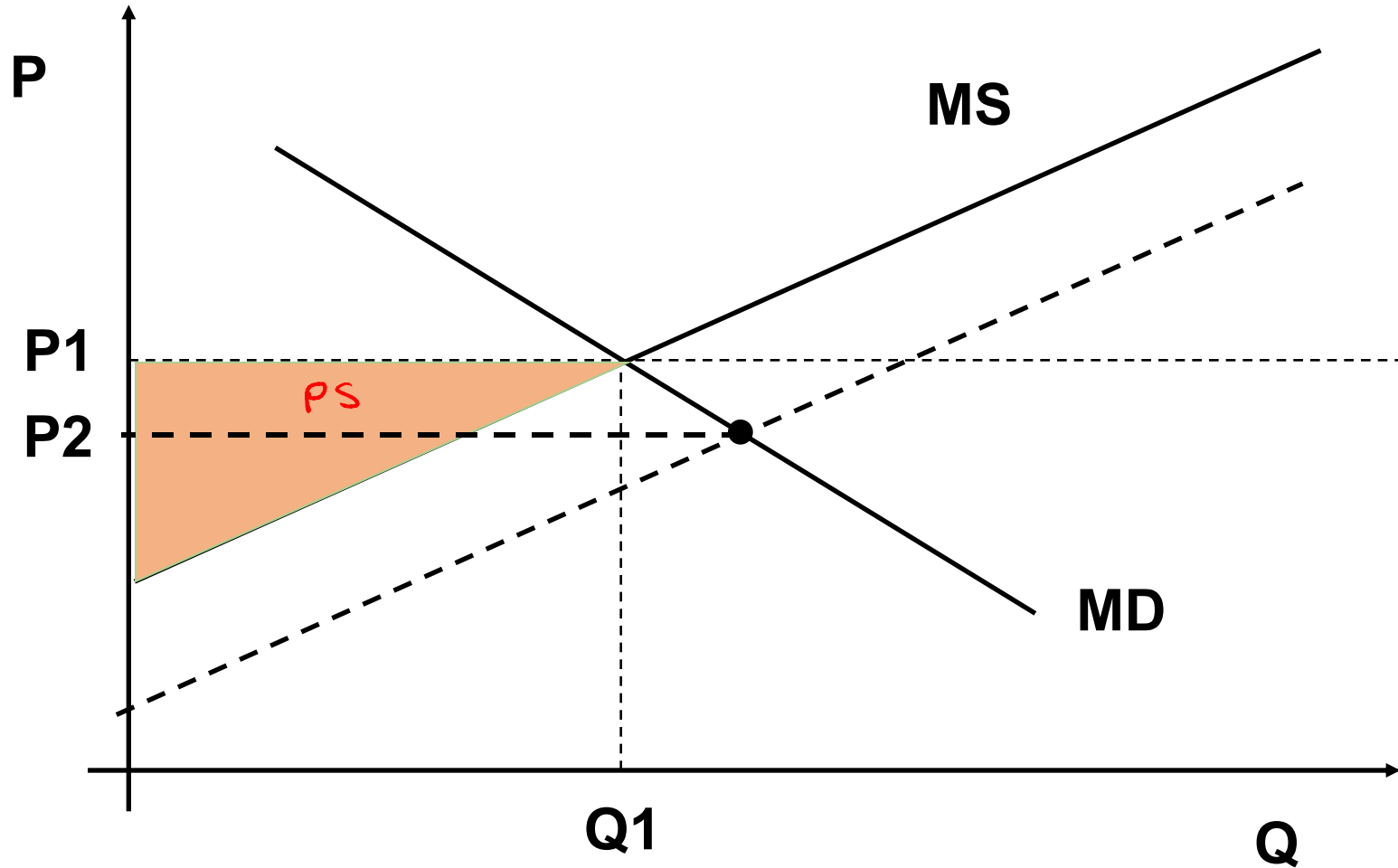


- Involve relationships among human beings and other living things and air, soil, and water
- Current concerns
 - Global warming
 - Loss of habitat and biodiversity
 - Air, water, and land pollution
- Responsibilities of firms
 - Eliminating toxic by-products of current manufacturing processes
 - Cleaning up prior environmental damage

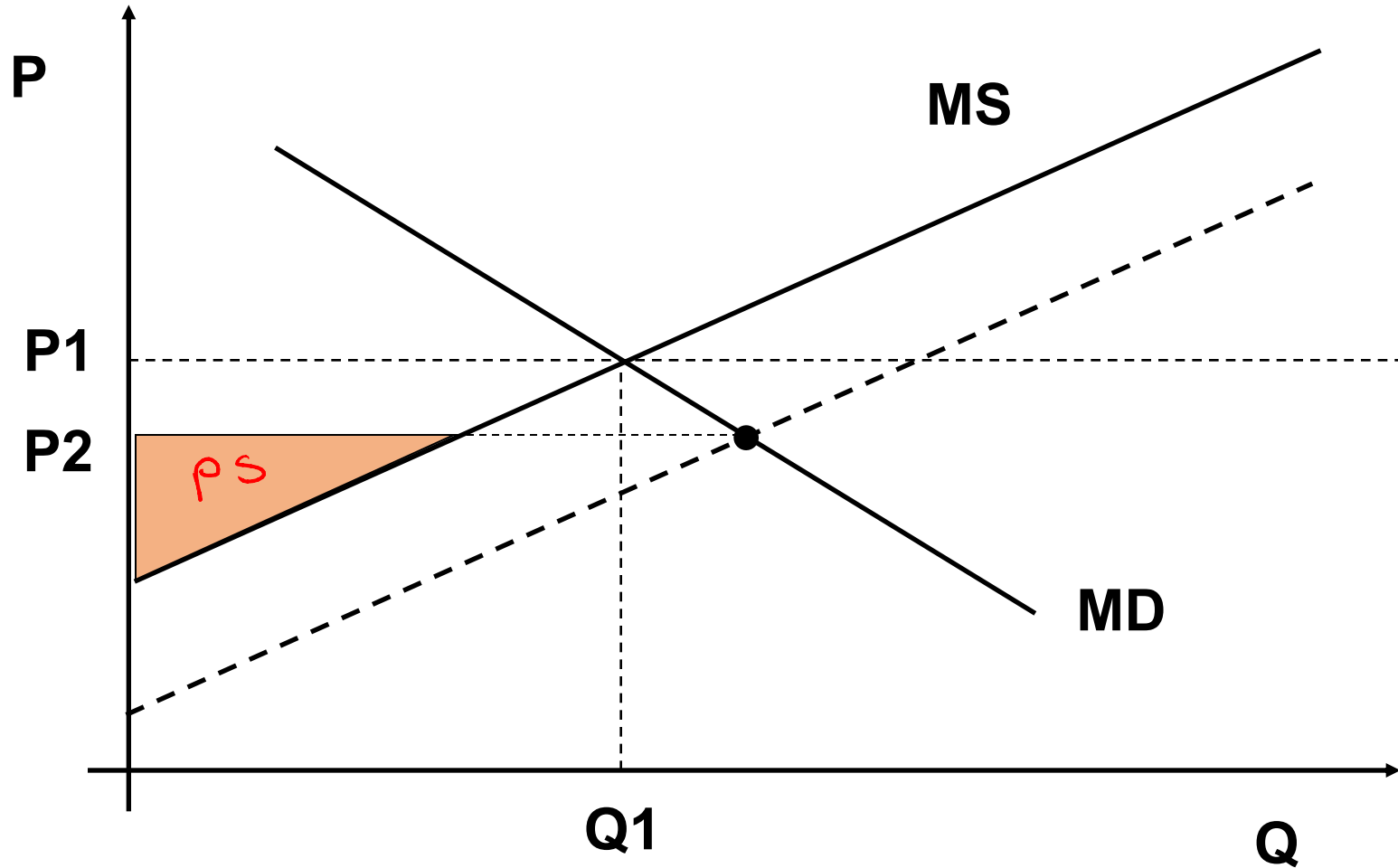
Porter's Five Forces Model



Threat of new entrants: Before...and After



New entrants erode incumbents producer surplus.



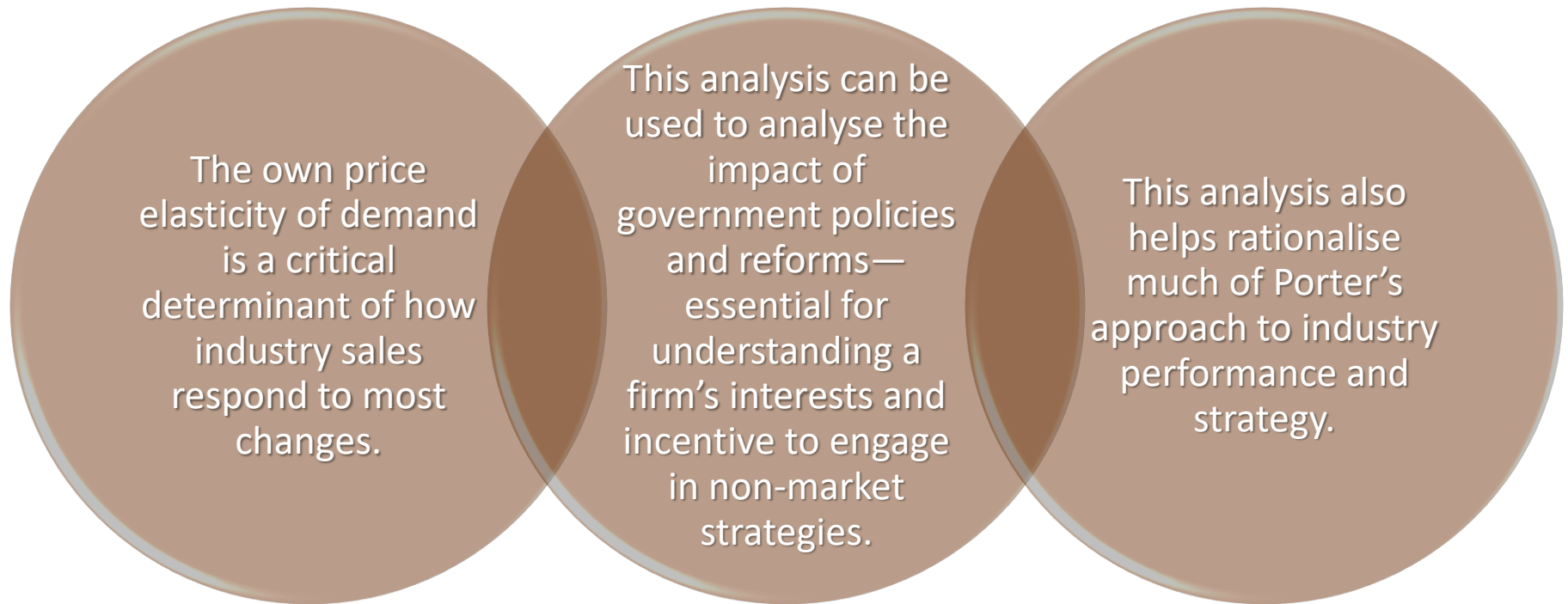
What about the other four forces?

Let's discuss each in turn



The straightforward supply-and-demand framework, when used to explain producer surplus, shows why four of the five forces are indeed threats to industry profitability.

Summary: Understanding a lot out from a straightforward framework.



Remember that supply-and-demand analysis tells you **how a competitive industry would respond**, not how an individual firm would respond to new circumstances.

Consider some background on McDonalds:

MCD: NYSE



Five Forces Driving Industry Competition

Determinants of Entry

Economies of scale **EoS**
 Proprietary product differences
 Brand identity
 Switching costs
 Capital requirements
 Access to distribution
 Absolute cost advantages
 Proprietary curve
 Access to necessary inputs
 Proprietary low-cost product design
 Government policy
 Expected retaliation

Determinants of Rivalry

Industry growth
 Fixed (or storage) costs/value added
 Intermittent overcapacity
 Product differences
 Brand identity
 Switching costs
 Concentration and balance
 Informational complexity
 Diversity of competitors
 Corporate stakes
 Exit barriers

Suppliers

Bargaining Power
of Suppliers

Bargaining Power
of Buyers

Buyers

Determinants of Supplier Power

Differentiation of inputs
 Switching costs of suppliers and firms
 in the industry
 Presence of substitute inputs
 Supplier concentration
 Importance of volume to supplier
 Cost relative to total purchases in
 the industry
 Impact of inputs on cost or differentiation
 Threat of forward integration relative to
 threat of backward integration by firms
 in the industry

Determinants of Buyer Power

Bargaining Leverage

Buyer concentration
 versus firm concentration
 Buyer volume
 Buyer switching costs
 relative to firm
 switching costs
 Buyer information
 Ability to backward
 integrate
 Substitute products
 Pull-through

Price Sensitivity

Price/total purchases
 Product differences
 Brand identity
 Impact on quality/
 performance
 Buyer profits
 Decision makers'
 incentives

**New
Entrants**

Threat of
New Entrants

**Industry
Competitors**

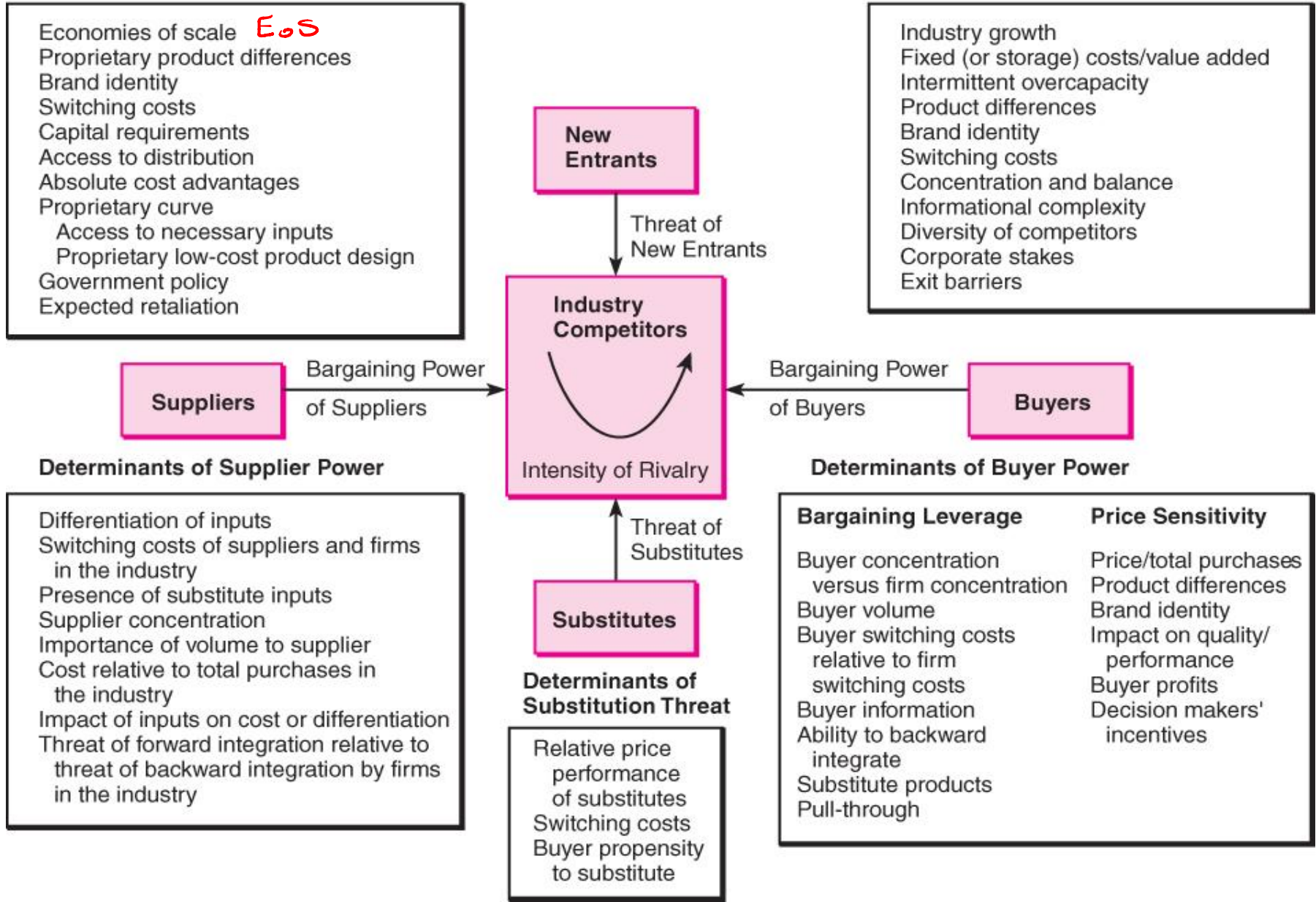
Intensity of Rivalry

Substitutes

Threat of
Substitutes

Determinants of Substitution Threat

Relative price
 performance
 of substitutes
 Switching costs
 Buyer propensity
 to substitute



Five Competitive Forces: McDonalds

COMPETITIVE RIVALRY WITHIN THE INDUSTRY

Five Competitive Forces: McDonalds

BARGAINING POWER OF SUPPLIERS

Five Competitive Forces: McDonalds

Five Competitive Forces: McDonalds

THREAT OF NEW ENTRANTS

Five Competitive Forces: McDonalds

THREAT OF SUBSTITUTE PRODUCTS

Five Competitive Forces: McDonalds

Porter Five Force	Intensity
Competitive rivalry within the industry	
Bargaining power of customers	
Threat of new entrants	
Bargaining power of suppliers	
Threat of substitute products	

A person is holding a red McDonald's french fry container filled with golden fries. The person's hand, with red-painted nails, is visible at the bottom left. The background is a dark, out-of-focus jacket. A large red rectangle is overlaid on the right side of the image, containing white text.

MCDONALD'S

& REAL ESTATE



THE NEXT 60 SECONDS MAY ALTER THE COURSE OF YOUR ENTIRE LIFE! *DON'T MISS A SINGLE WORD!*

YOU ARE LOOKING AT THE
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*The most revolutionary
development in the Restau-
rant industry during the past
50 years!*

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SELF SERVICE!**

NO

CAR HOOPS
WAITRESSES
DISH WASHERS
BUS BOYS
GLASSWARE
DISHES
SILVERWARE
WAITERS

**The McDonald system
eliminates all of this!**

IN OUR MODEL DRIVE-IN, HERE IN SAN
BERNARDINO, CALIF., WE SELL . . .

MORE THAN A MILLION HAMBURGERS
A YEAR!

MORE THAN A HALF MILLION MALTS,
SHAKES, AND SOFT DRINKS A YEAR!

MORE THAN A HALF MILLION ORDERS
OF FRENCH FRIES A YEAR!

- Let's hear from Dr Porter...
- https://www.youtube.com/watch?v=mYF2_FBCvXw&feature=results_video&playnext=1&list=PL72A65662D7647704

TED Talk – Drug Cartels – Do you see the Economics?

- <https://www.youtube.com/watch?v=VYU25aJpg5o>